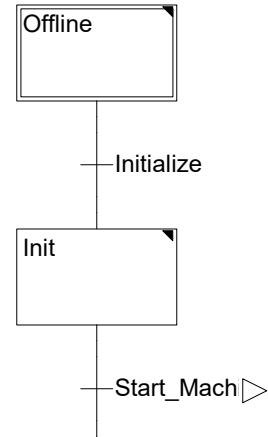
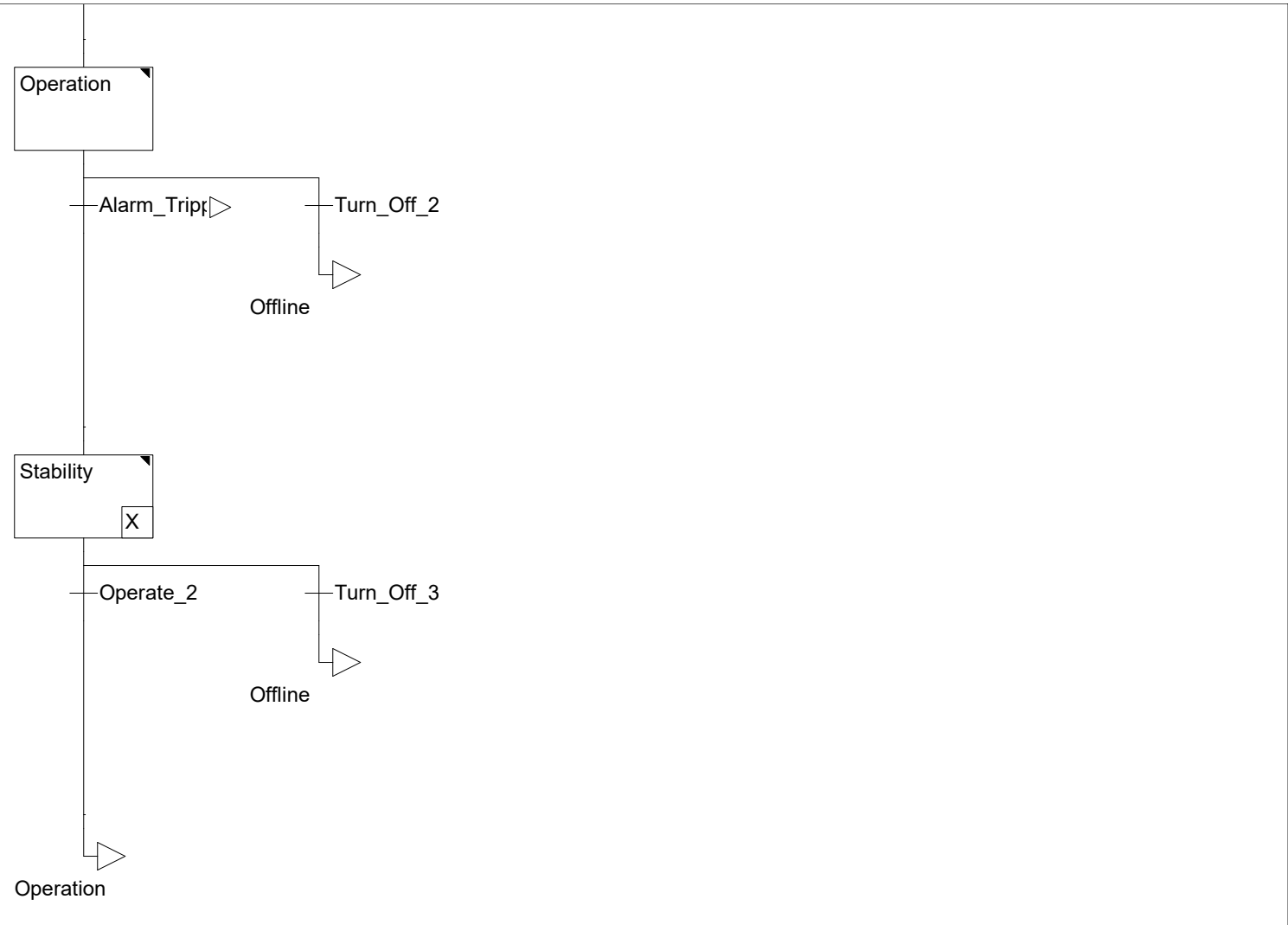


Main (PRG-SFC)

0001	PROGRAM Main
0002	VAR
0003	count: DINT;
0004	countreset: BOOL;
0005	IncrementTimer: TON;
0006	ONS: R_TRIG;
0007	OffDelay: TON;
0008	OffTrigger: BOOL;
0009	LLLvDelay: TON;
0010	HHLvDelay: TON;
0011	TempHDelay: TON;
0012	TempLDelay: TON;
0013	
0014	BOOLLH: BOOL;
0015	BOOLLL: BOOL;
0016	BOOLTH: BOOL;
0017	BOOLM: BOOL;
0018	PreAlarmBOOL : BOOL;
0019	MixerHDelay: TON;
0020	MixerDelay: TON;
0021	BOOL1: BOOL;
0022	BOOLVL: BOOL;
0023	VLLvDelay: TON;
0024	Test1: BOOL;
0025	Initialize: BOOL;
0026	Start_Machine: BOOL;
0027	Turn_Off: BOOL;
0028	Turn_Off_2: BOOL;
0029	Turn_Off_3: BOOL;
0030	Alarm_Tripped: BOOL;
0031	Alarm_Tripped_2: BOOL;
0032	Operate: BOOL;
0033	Operate_2: BOOL;
0034	BOOLTL: BOOL;
0035	END_VAR





Main (PRG-SFC).Action Offline (LD)

0001	Offline Mode (Mode 0): After 5 seconds, resets pump, heater, mixer, and valve. Transitions to Init stage when on button is pressed and machine
	Turn_Off (R)
	Turn_Off_2 (R)
	Turn_Off_3 (R)
	LL_Lvl_Notify (R)

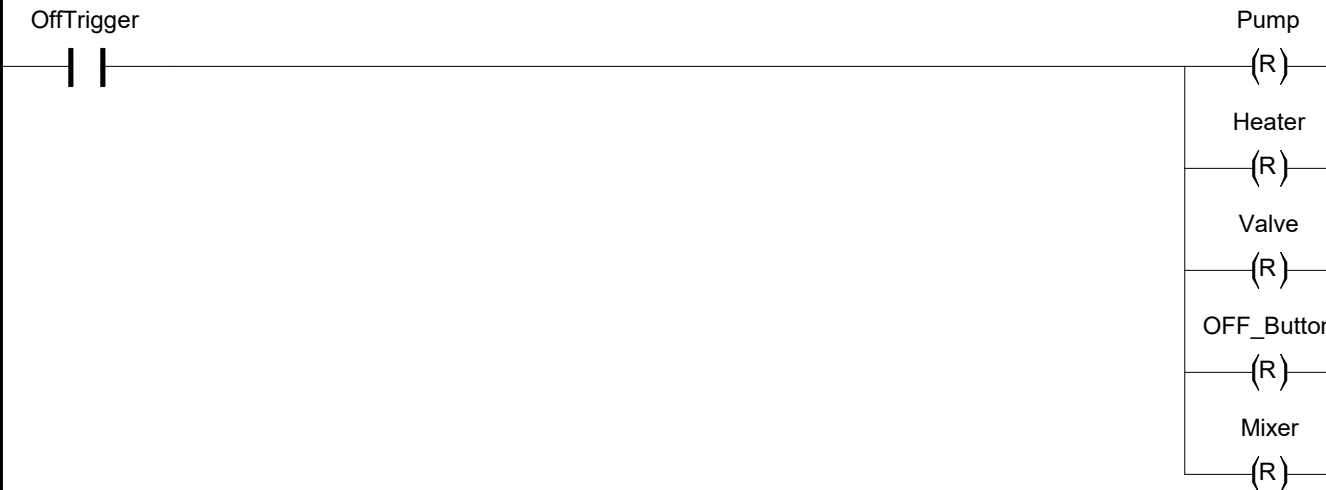
0002



0003



0004

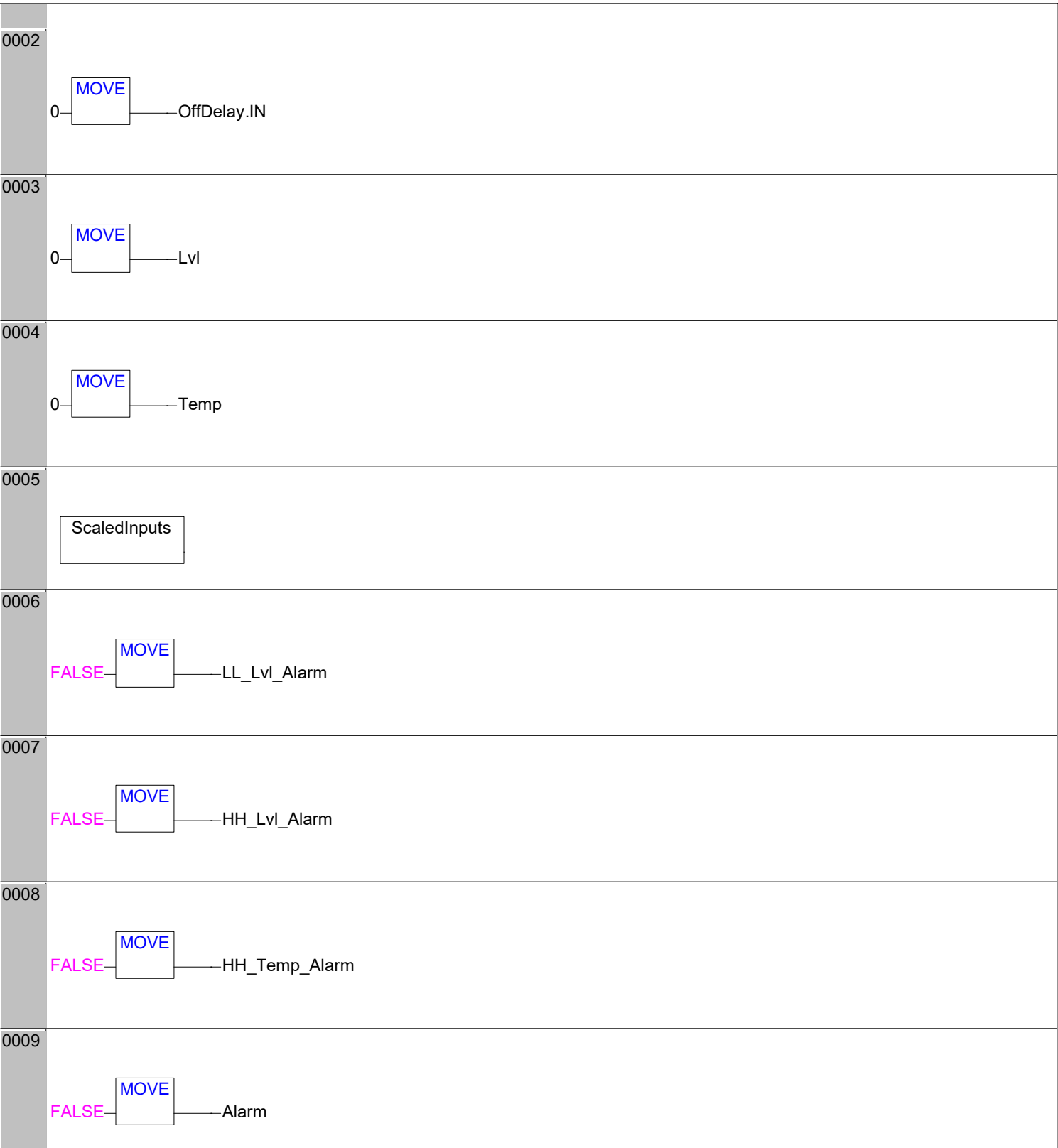


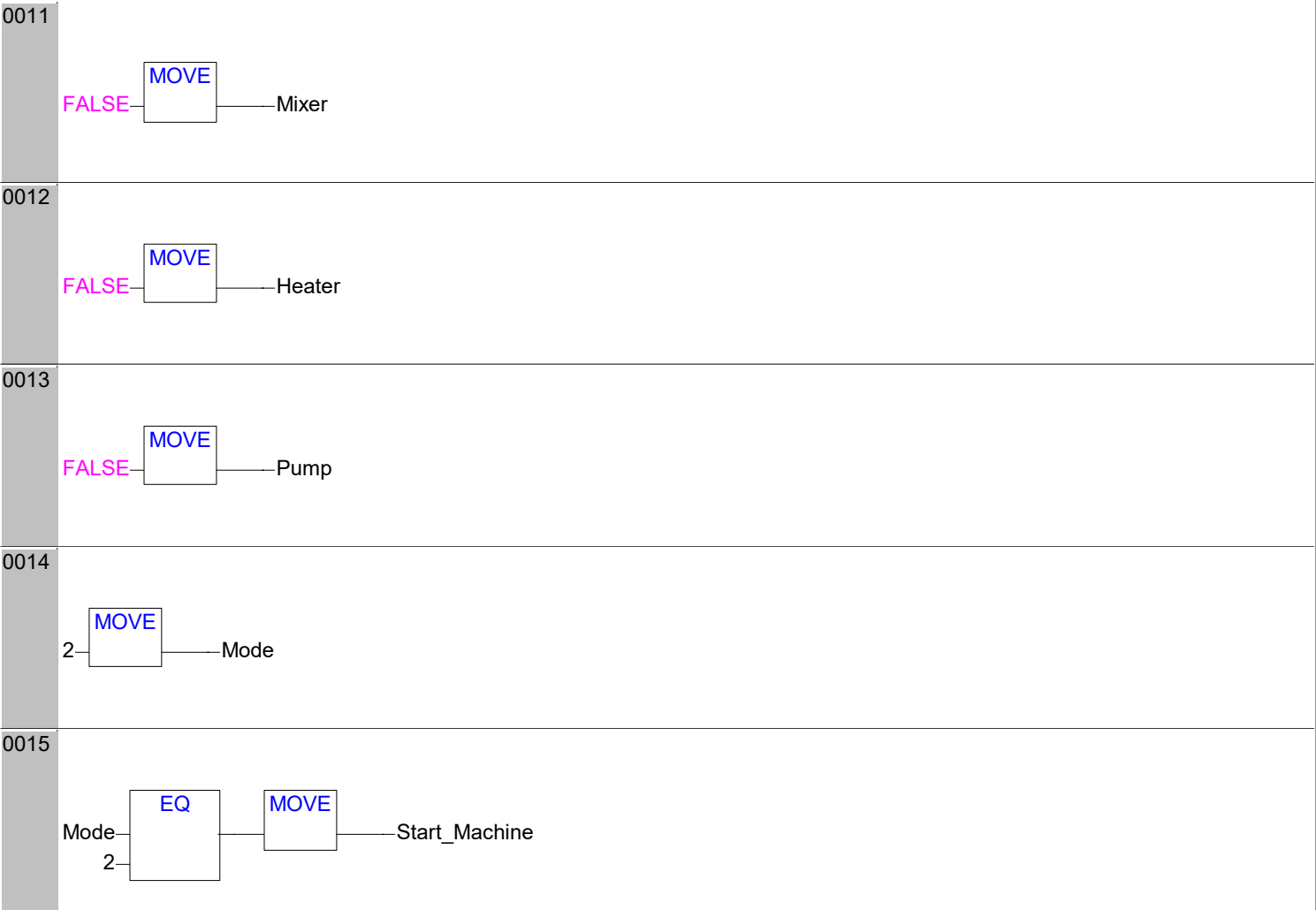
0005



0006







Main (PRG-SFC).Action Startup - Entry (ST)

0001	(*
0002	Count how many times startup mode is entered.
0003	*)
0004	count := count + 1;
0005	
0006	IF countreset = TRUE THEN
0007	count :=0;
0008	countreset := FALSE;
0009	END_IF;

Main (PRG-SFC).Action Startup (ST)

0001	(*
0002	Startup Mode: Turns pump on until range within 40-70. Then turn heater on until heater between 300-350 degrees. Then turn mixer on for at least 30
0003	seconds. After 30 seconds, transition to Operation Mode.
0004	*)
0005	Start_Machine := 0;
0006	IF PreAlarmBOOL = TRUE THEN

0019	IF LvlScale > 20 THEN
0020	PreAlarmBOOL := TRUE;
0021	END_IF;
0022	HHLvlDelay(IN :=BOOLLH, PT:= T#5s);
0023	IF LvlScale >70 THEN
0024	BOOLLH :=1;
0025	IF HHLvlDelay.Q = TRUE THEN
0026	Pump :=0;
0027	Valve:=1;
0028	END_IF;
0029	END_IF;
0030	VLLvlDelay(IN :=BOOLVL, PT:= T#5s);
0031	IF LvlScale <70 AND LvlScale >60 THEN
0032	BOOLVL :=1;
0033	IF VLLvlDelay.Q = TRUE THEN
0034	Valve :=0;
0035	END_IF;
0036	END_IF;
0037	
0038	TempHDelay(IN :=BOOLTH, PT:= T#5s);
0039	IF LvlScale>40 AND LvlScale<70 AND TempScale < 300 THEN
0040	BOOLTH:=1;
0041	IF TempHDelay.Q = TRUE THEN
0042	Heater :=1;
0043	END_IF;
0044	END_IF;
0045	
0046	TempLDelay(IN :=BOOLTL, PT:= T#5s);
0047	IF TempScale > 350 THEN
0048	BOOLTL:=1;
0049	IF TempLDelay.Q = TRUE THEN
0050	Heater :=0;
0051	END_IF;
0052	END_IF;
0053	
0054	MixerHDelay(IN :=BOOLM, PT:= T#5s);
0055	MixerDelay(IN :=BOOL1, PT:= T#30s);
0056	IF TempScale >300 AND TempScale < 350 THEN
0057	BOOLM:=1;
0058	IF MixerHDelay.Q = TRUE THEN
0059	Mixer :=1;
0060	END_IF;
0061	BOOL1:= TRUE;
0062	END_IF;
0063	
0064	IF MixerDelay.Q AND TempScale <300 THEN
0065	Mode :=3;
0066	BOOL1:= 0;
0067	END_IF;
0068	(*

0081	
0082	IF HH_Lvl_Trigger = TRUE AND Alarm_Reset_Button =FALSE AND HH_Lvl_Release = FALSE THEN
0083	HH_Lvl_Alarm := TRUE;
0084	HH_Lvl_Notify := TRUE;
0085	HH_Lvl_Trigger:= FALSE;
0086	END_IF;
0087	
0088	IF HH_Temp_Trigger = TRUE AND Alarm_Reset_Button =FALSE AND HH_Temp_Release = FALSE THEN
0089	HH_Temp_Alarm := TRUE;
0090	HH_Temp_Trigger := FALSE;
0091	HH_Temp_Notify := TRUE;
0092	END_IF;
0093	
0094	IF LL_Lvl_Alarm OR HH_Lvl_Alarm OR HH_Temp_Alarm THEN
0095	Alarm :=1;
0096	END_IF;
0097	
0098	IF Alarm = 1 THEN
0099	Mode :=4;
0100	END_IF;
0101	
0102	IF Mode <> 2 THEN
0103	PreAlarmBOOL := FALSE;
0104	BOOLL_L :=0;
0105	BOOLL_H :=0;
0106	BOOLL_VL :=0;
0107	BOOLL_TH:=0;
0108	BOOLL_TL:=0;
0109	BOOLL_M:=0;
0110	END_IF;
0111	
0112	IF Mode = 3 THEN
0113	Operate:= TRUE;
0114	END_IF;
0115	
0116	IF Mode = 4 THEN
0117	Alarm_Tripped:= TRUE;
0118	END_IF;
0119	
0120	IF Mode = 0 THEN
0121	Turn_Off:= TRUE;
0122	END_IF;
Main (PRG-SFC).Action Operation (ST)	
0001	(*
0002	Operation Mode: Call Operation1 to start Operation Mode actions. Simulation simulates level and temp changes, Scaled Inputs scales level and
0003	temp inputs.
0004	Alarm Trigger checks for conditions to trigger alarm and switch to stability mode. Transition to Stability Mode if mode is 4. Transtion to Offline Mode if
0005	*)
0006	
0007	Operate :=FAI SF;

0019	Off_Button := FALSE;
0020	Turn_Off_2 :=1;
0021	END_IF;

Main (PRG-SFC).Action Stability (IL)

0001	(*
0002	If Low level alarm is on, turn on pump and close valve until alarm turns off. If High level alarm is on,
0003	close pump and turn on valve until alarm turns off.
0004	If high temp alarm is on, turn off heater.
0005	
0006	Keeps track of how much time spent in stability mode.
0007	Transition to Operation Mode when alarms are cleared. Transtion to Offline Mode if OFF button or Estop is pressed.
0008	*)
0009	LD FALSE
0010	ST Alarm_Tripped_2
0011	LD FALSE
0012	ST Alarm_Tripped
0013	LD Alarm
0014	JMPCN TransMode3
0015	LD E_Stop
0016	JMPCN TransMode0
0017	LD OFF_Button
0018	JMPC TransMode0
0019	CAL Stability1
0020	CAL Simulation
0021	CAL ScaledInputs
0022	CAL AlarmRelease
0023	LD IncrementTimer.Q
0024	JMPCN Timer
0025	LD IncrementTimer.Q
0026	JMPC notimer
0027	JMP EndofLine
0028	
0029	TransMode0:
0030	LD 0
0031	ST Mode
0032	JMP Transition
0033	RET
0034	
0035	TransMode3:
0036	LD 3
0037	ST Mode
0038	JMP Transition
0039	RET
0040	
0041	Transition:
0042	LD Mode
0043	EQ 3
0044	JMPC OperationTrans
0045	LD Mode
0046	EQ 0

0058	LD	FALSE
0059	ST	Alarm_Reset_Button
0060	LD	TRUE
0061	ST	Turn_Off_3
0062	RET	
0063		
0064		
0065	Timer:	
0066	LD	T#1000ms
0067	ST	IncrementTimer.PT
0068	LD	TRUE
0069	ST	IncrementTimer.IN
0070	CAL	IncrementTimer
0071	LD	IncrementTimer.Q
0072	JMPC	AddSec
0073	RET	
0074		
0075	AddSec:	
0076	LD	Sec
0077	ADD	1
0078	ST	Sec
0079	LD	Sec
0080	EQ	60
0081	JMPC	AddMin
0082	RET	
0083		
0084	AddMin:	
0085	LD	Minu
0086	ADD	1
0087	ST	Minu
0088	LD	0
0089	ST	Sec
0090	LD	Minu
0091	EQ	60
0092	JMPC	AddHour
0093	RET	
0094		
0095	AddHour:	
0096	LD	H
0097	ADD	1
0098	ST	H
0099	LD	0
0100	ST	Minu
0101	RET	
0102		
0103	notimer:	
0104	LD	FALSE
0105	ST	IncrementTimer.IN
0106	CAL	IncrementTimer
0107	RET	

0007
0008
0009
0010
0011

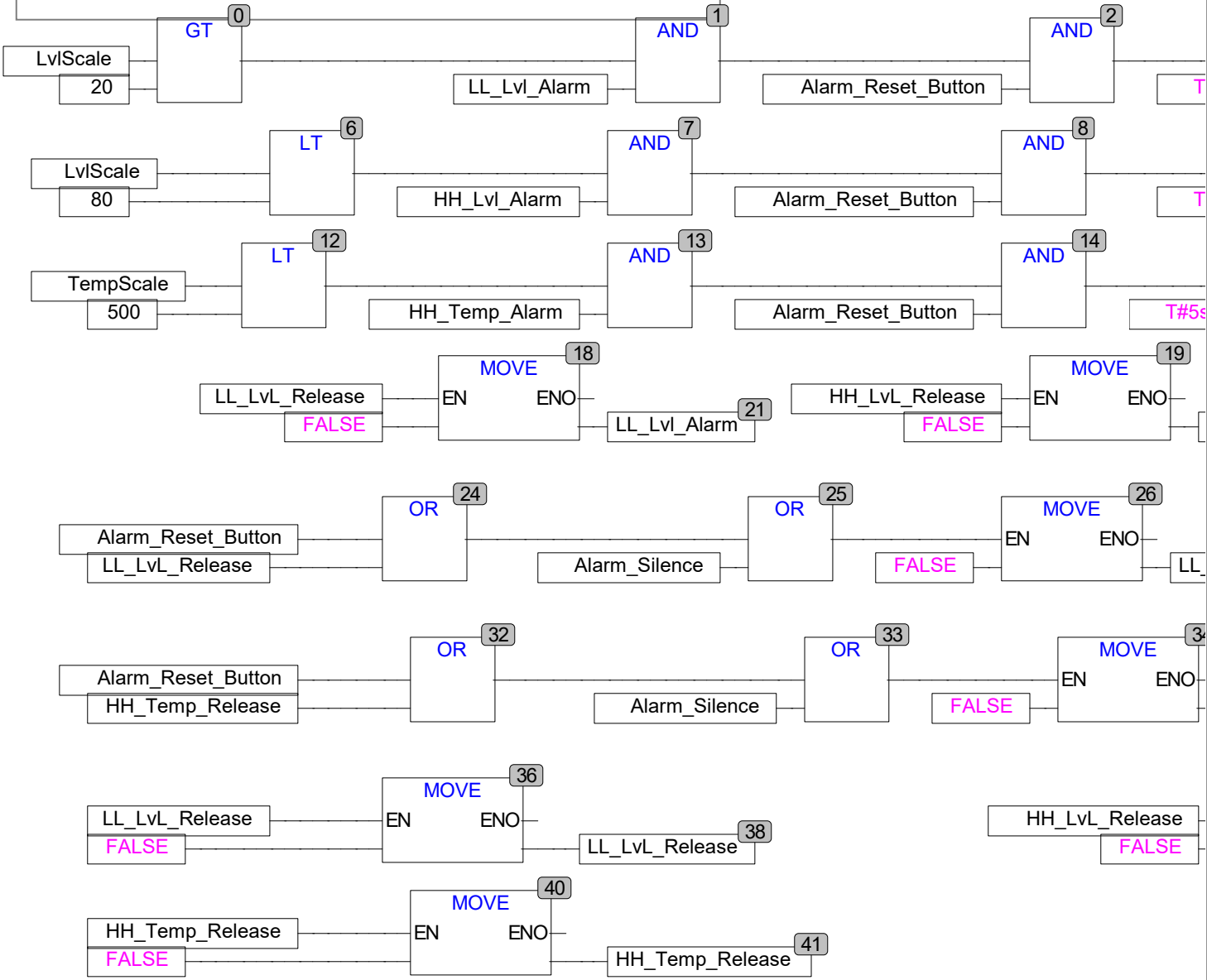
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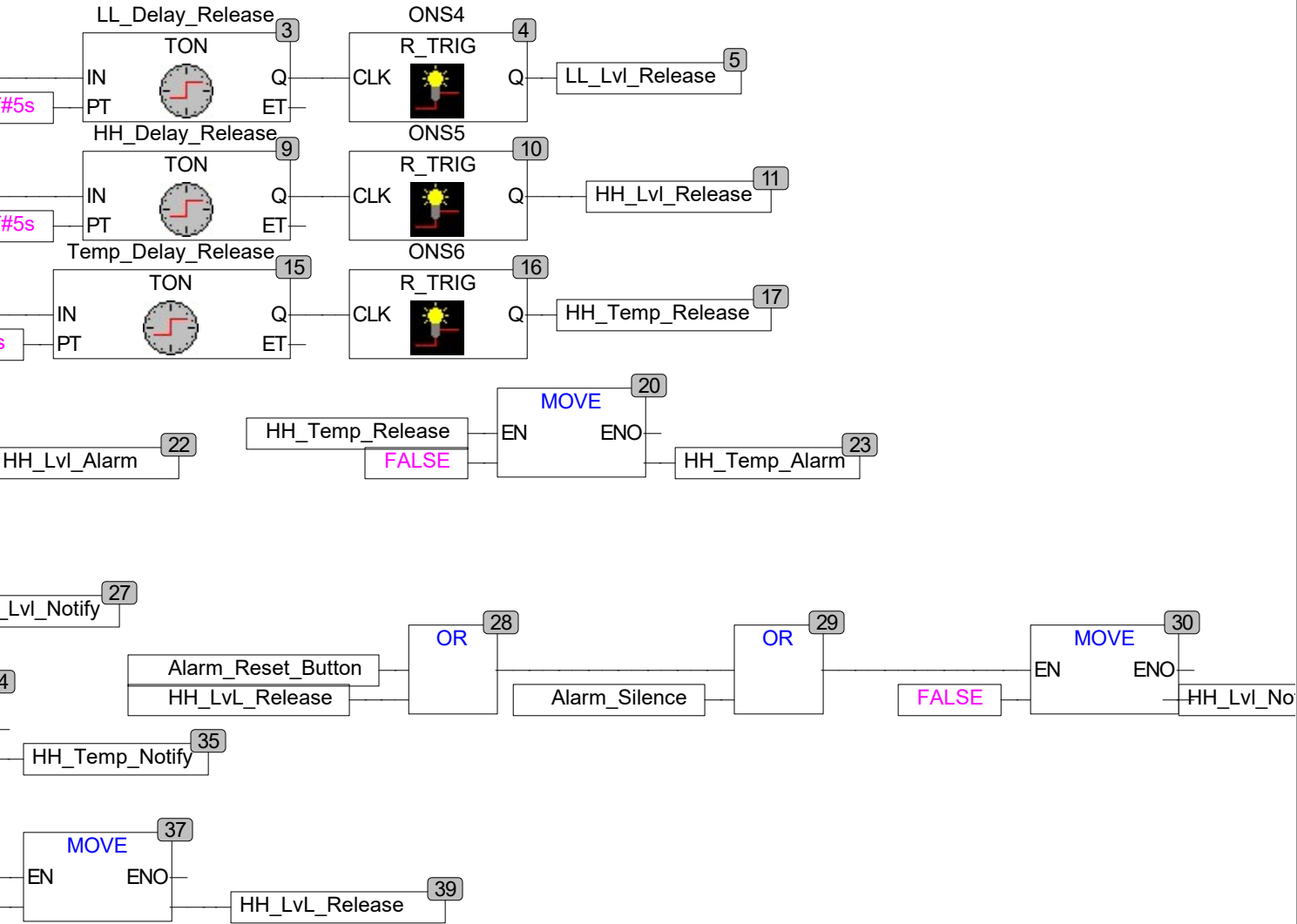
Seconds
Minu
Minutes
H
Hours

AlarmRelease (PRG-CFC)

```
0001 PROGRAM AlarmRelease
0002 VAR
0003     LL_Delay_Release: TON;
0004     HH_Delay_Release: TON;
0005     ONS5: R_TRIG;
0006     ONS6: R_TRIG;
0007     ONS4: R_TRIG;
0008     Temp_Delay_Release: TON;
0009 END_VAR
```

If alarm conditions are no longer met and alarms are still on for longer than 5 seconds, turn off alarms and notifications.







0007

ONS3: R_TRIG;

0008

LL_Delay_Trigger: TON;

0009

HH_Delay_Trigger: TON;

0010

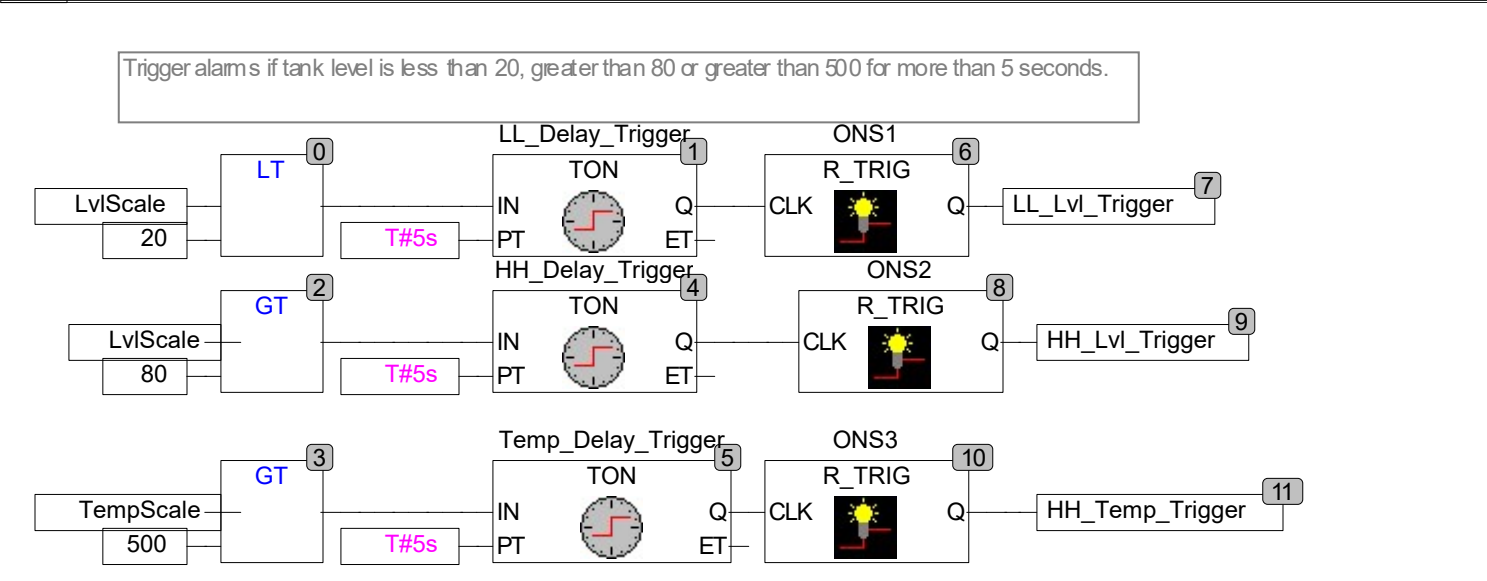
Temp_Delay_Trigger: TON;

0011

ONS1: R_TRIG;

0012

END_VAR



ScaledInputs (PRG-CFC)

0001

PROGRAM ScaledInputs

0002

VAR

0003

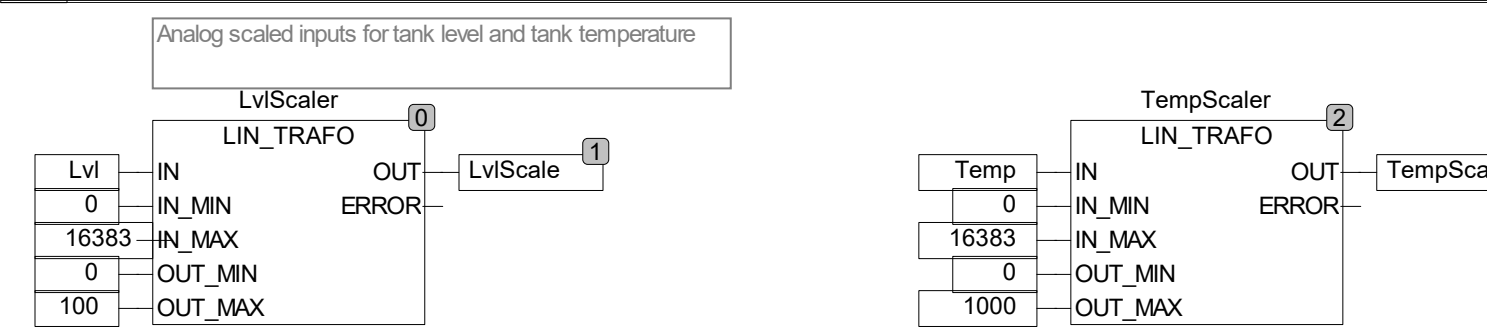
TempScaler: LIN_TRAFO;

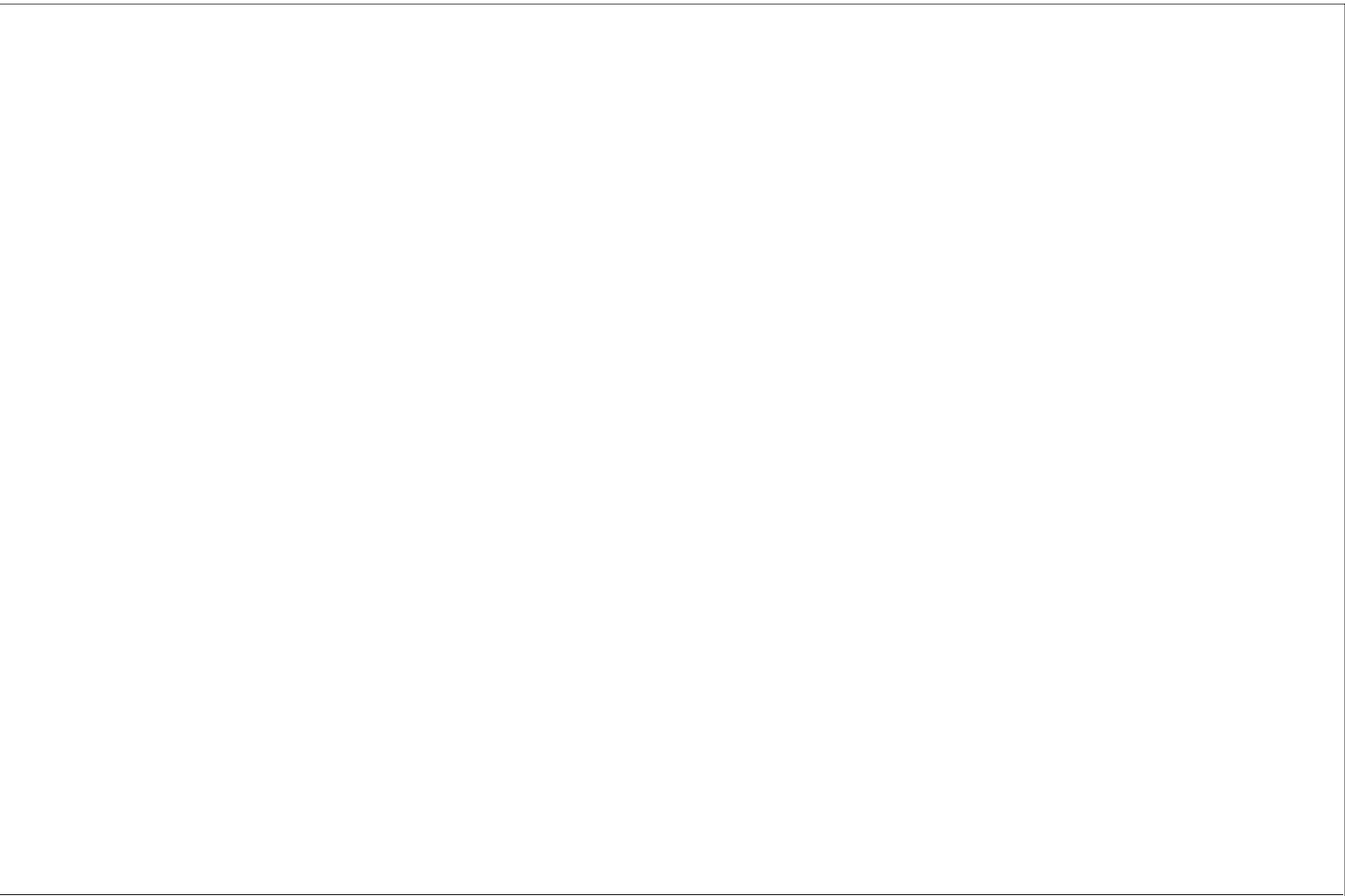
0004

LvlScaler: LIN_TRAFO;

0005

END_VAR





le

3

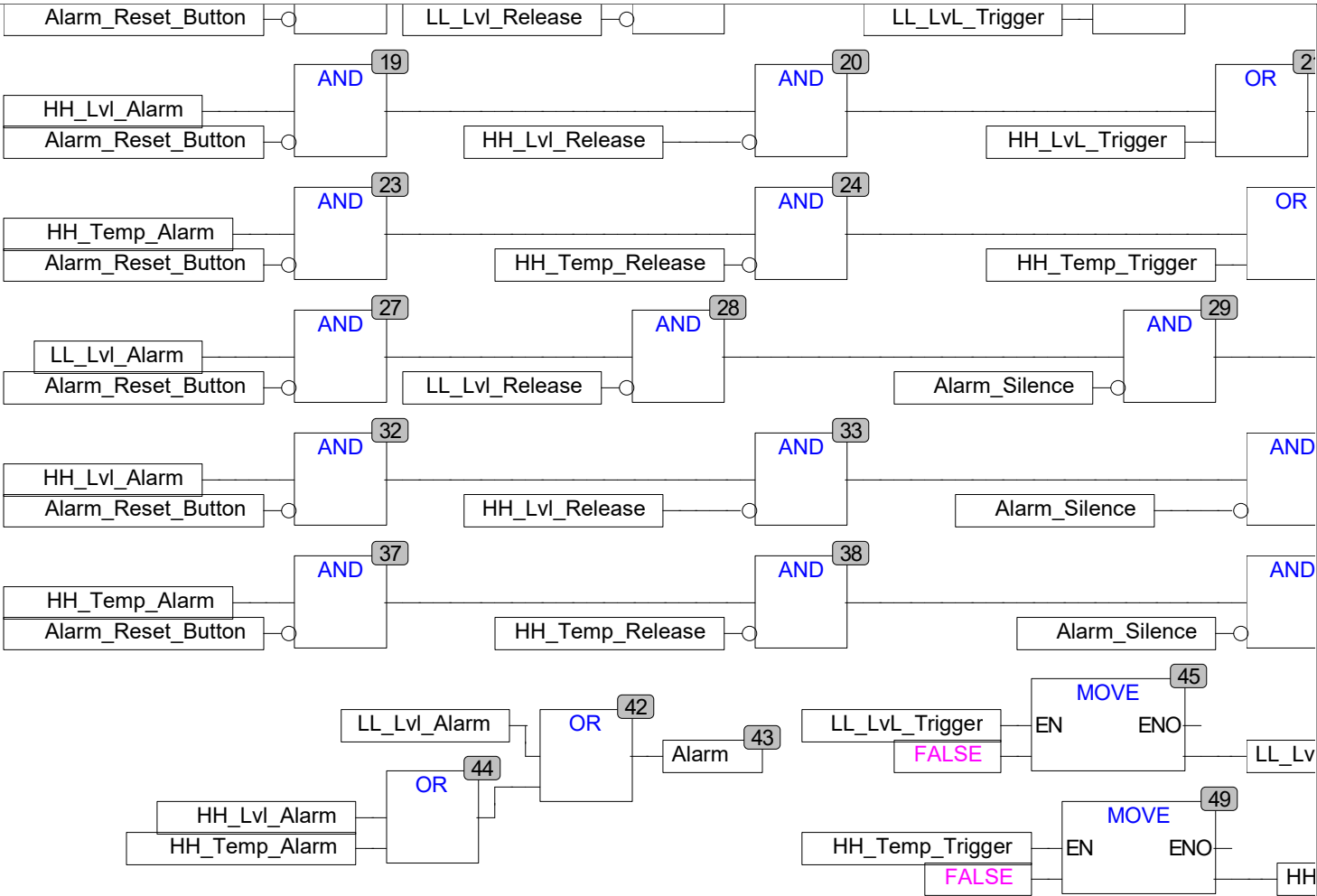
Operation1 (PRG-CFC)

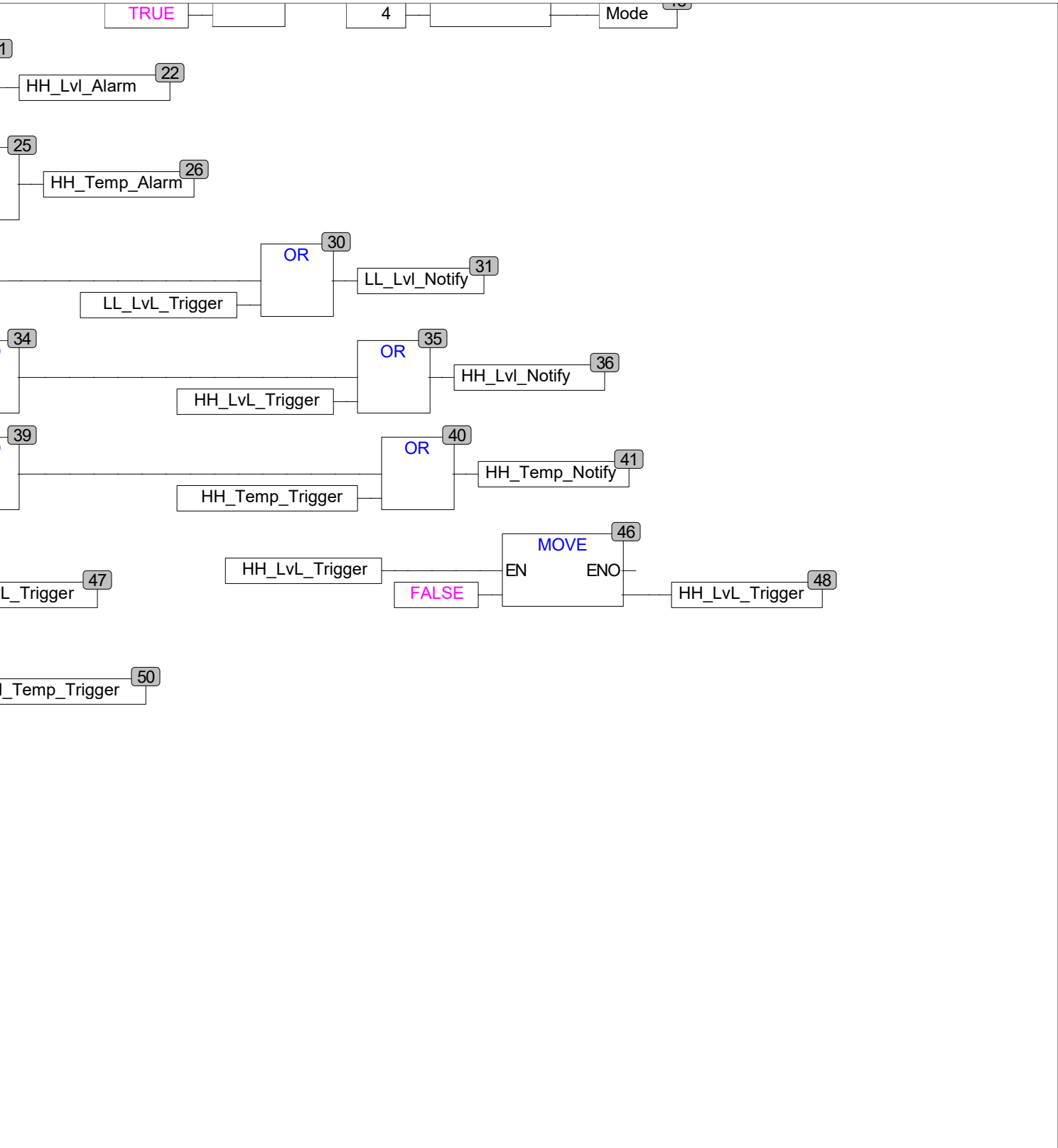
0001	PROGRAM Operation1
0002	VAR
0003	HeaterDelay: TON;
0004	ValveDelay: TON;
0005	END_VAR





han 400 degrees, turn heater on. Change mode to 4 if alarms are triggered.





Simulation (PRG-ST)

0001	PROGRAM Simulation
0002	VAR
0003	HEATUP: DINT;
0004	DROPHEAT: DINT;
0005	FILLRATE: DINT;
0006	DRAINRATE: DINT;
0007	TONDelay: TON;
0008	DelayBOOL: BOOL;
0009	END_VAR
0001	(*
0002	Simulate tank temperature and tank level changes. Values update every 5 seconds while any device is on. Temp slowly falls to 70.
0003	Pump raises level and lowers temperature, valve lowers level, and heater raises temperature.
0004	*)
0005	HEATUP :=1500;
0006	DROPHEAT :=500;
0007	FILLRATE :=1500;
0008	DRAINRATE :=400;
0009	

0021	END_IF;
0022	IF Temp > 500 THEN
0023	Temp := Temp - DROPHEAT;
0024	END_IF;
0025	ScaledInputs;
0026	END_IF;
0027	
0028	IF VALVE = TRUE THEN
0029	IF TONDelay.Q = TRUE AND Lvl > 400 THEN
0030	Lvl := Lvl - DRAINRATE;
0031	ScaledInputs;
0032	END_IF;
0033	END_IF;
0034	
0035	IF HEATER = TRUE THEN
0036	IF TONDelay.Q = TRUE AND Temp < 14500 THEN
0037	Temp := Temp + HEATUP;
0038	ScaledInputs;
0039	END_IF;
0040	END_IF;
0041	END_IF;
0042	IF TONDelay.Q = TRUE THEN
0043	DelayBool := FALSE;
0044	END_IF;
0045	RETURN;

Stability1 (PRG-IL)

0001	PROGRAM Stability1
0002	VAR
0003	LL_Timer: TON;
0004	HH_Timer: TON;
0005	Temp_Timer: TON;
0006	END_VAR

0001	(*
0002	If Low level alarm is on, turn on pump and close valve until alarm turns off. If High level alarm is on, close pump and turn on valve until alarm turns off
0003	If high temp alarm is on, turn off heater.
0004	*)
0005	LD Mode
0006	EQ 4
0007	AND LL_Lvl_Alarm
0008	JMPC LL_Timer
0009	LD Mode
0010	EQ 4
0011	AND HH_Lvl_Alarm
0012	JMPC HH_Lvl_Timer
0013	LD Mode
0014	EQ 4
0015	AND HH_Temp_Alarm
0016	JMPC HH_Temp_Timer
0017	LD Mode
0018	NE 4

0030	RET
0031	
0032	LL_Lvl:
0033	LD TRUE
0034	ST Pump
0035	LD FALSE
0036	ST VALVE
0037	RET
0038	
0039	HH_Lvl_Timer:
0040	LD T#10s
0041	ST HH_Timer.PT
0042	LD TRUE
0043	ST HH_Timer.IN
0044	CAL HH_Timer
0045	LD HH_Timer.Q
0046	JMPC HH_Lvl
0047	RET
0048	
0049	HH_Lvl:
0050	LD FALSE
0051	ST Pump
0052	LD TRUE
0053	ST VALVE
0054	RET
0055	
0056	HH_Temp_Timer:
0057	LD T#10s
0058	ST Temp_Timer.PT
0059	LD TRUE
0060	ST Temp_Timer.IN
0061	CAL Temp_Timer
0062	LD Temp_Timer.Q
0063	JMPC HH_Temp
0064	RET
0065	
0066	HH_Temp:
0067	LD FALSE
0068	ST Heater
0069	LD 0
0070	ST Temp_Timer.IN
0071	RET
0072	
0073	StabilityTimerReset
0074	LD 0
0075	ST LL_Timer.IN
0076	CAL LL_Timer
0077	LD 0
0078	ST HH_Timer.IN
0079	CAL HH_Timer
0080	LD 0

